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DETAIL SPECIFICATION

CLOTH, DUCK, COTTON OR COTTON- POLYESTER BLEND, SYNTHETIC RUBBER, IMPREGNATED, AND LAMINATED, OIL RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers a laminated cotton duck or cotton-polyester blend duck cloth that has been impregnated with an oil resistant synthetic rubber.

1.2 Classification. The cloth is available in the following types as specified (see 6.2).

1.2.1 Types.

Type I - Sheets
Type II - Strips
Type III - Cut items
Type IV - Molded items

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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AMSC N/A FSC 8305

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. N/A

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

Title16 CFR Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM20A - Test Method for Fiber Analysis: Quantitative
AATCC TM30 - Test Method for Antifungal Activity, Assessment of Textile Materials: Mildew and Rot Resistance of Textile Materials

(Copies of these documents are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASSOCIATION FOR RUBBER PRODUCTS MANUFACTURERS

ARPM MO-1 – Handbook for Molded, Extruded, Lathe Cut and Cellular Products

(Copies of this document are available online at https://www.arpminc.com.)

ASTM INTERNATIONAL

ASTM D471 - Standard Test Method for Rubber Property- Effect of Liquids
ASTM D792 - Standard Test Methods for Density and Specific Gravity (Relative Density of Plastics by Displacement)

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ASTM D1777	- Standard Test Method for Thickness of Textile Materials
ASTM D3775	- Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M	- Standard Test Methods for Mass per Unit Area (Weight) of Fabric
ASTM D5034	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise indicated, the finished cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Duck yarn. The duck shall be made of high quality cotton or a 50/50 cotton-polyester blend.

3.4.2 Synthetic rubber. The synthetic rubber used for impregnating/laminating the cotton or cotton-polyester blend duck material shall be a compound which shall conform to the requirements of this specification.

3.4.3 Weave. The cotton cloth shall be a plain weave made with 2-ply warp and filling yarns and the cotton-polyester blend cloth shall be made with single yarns.

3.5 Physical requirements. The impregnated/laminated material shall conform to the following physical requirements as stated in Table I, when tested as specified in 4.5.

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TABLE I. Physical requirements.

Test	Requirement			
	Type I	Type II	Type III	Type III
Weight, oz/yd² (min.)	8.0	8.0	8.0	8.0
Yarns per inch Warp Filling	 50 (± 1) 40 (± 2)	 50 (± 1) 40 (± 2)	 50 (± 1) 40 (± 2)	 50 (± 1) 40 (± 2)
Breaking strength, lbs per inch per width (min.) Warp Filling	 150 140	 150 140	 150 140	 150 140
Thickness, inches, tolerance	± 0.031	± 0.031	± 0.031	± 0.031
Width, tolerance, inches	-	1/4-1/2 (inclusive) (± 1/32) 1/2-1 (inclusive) (± 3/64) Over 1 (± 1/16)	-	-
Density, lb/ft³ (min.)	67	67	67	67
Permanent set, percent, (max) Compressive stress @ lb/in² 500 1000 3000 5000 10000	 3.0 4.0 7.0 10.0 13.0	 3.0 4.0 7.0 10.0 13.0	 3.0 4.0 7.0 10.0 13.0	 3.0 4.0 7.0 10.0 13.0
Liquid resistance: Change in volume (swell), percent (max) Delamination	 25 No delamination	 25 No delamination	 25 No delamination	 25 No delamination
Fungus (max)	Microscopic	Microscopic	Microscopic	Microscopic

3.5.1 Age. The age of the impregnated cloth based on the month in which it is cure shall not exceed 12 months at the time of acceptance under contract by the purchaser. Material shall be rejected when the cure date cannot be determined.

3.5.2 Load deflection. The material shall be within the deflection limits shown on Table II, when tested as specified in 4.5.

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TABLE II. Load deflection.

	Deflections of laminated material											
	Thickness (inches)											
Load, lb/in²	14/64 – 16/64 inch thick		17/64 – 19/64 inch thick		21/64-23/64 inch thick		15/32 -17/32 inch thick		19/32 – 21/32 inch thick		61/64-67/64 inch thick	
	Min. (in.)	Max (in.)	Min. (in.)	Max (in.)	Min. (in.)	Max (in.)	Min. (in.)	Max (in.)	Min. (in.)	Max (in.)	Min. (in.)	Max (in.)
52	0.001	0.005	0.002	0.005	0.002	0.007	0.003	0.009	0.004	0.011	0.006	0.019
100	0.002	0.007	0.003	0.009	0.004	0.011	0.005	0.014	0.007	0.017	0.010	0.028
200	0.003	0.011	0.006	0.013	0.007	0.016	0.009	0.021	0.013	0.027	0.018	0.043
300	0.005	0.014	0.008	0.017	0.010	0.020	0.013	0.027	0.018	0.034	0.025	0.054
400	0.006	0.016	0.010	0.020	0.012	0.024	0.016	0.033	0.022	0.040	0.031	0.064
500	0.008	0.018	0.012	0.023	0.014	0.027	0.019	0.038	0.026	0.046	0.037	0.073
600	0.009	0.020	0.013	0.025	0.016	0.030	0.022	0.042	0.029	0.052	0.043	0.081
700	0.010	0.022	0.015	0.027	0.018	0.033	0.025	0.047	0.033	0.056	0.049	0.090
800	0.012	0.024	0.017	0.029	0.019	0.035	0.028	0.051	0.036	0.062	0.055	0.098
900	0.013	0.025	0.018	0.031	0.021	0.038	0.031	0.055	0.040	0.066	0.060	0.105
1000	0.014	0.027	0.020	0.033	0.023	0.040	0.034	0.058	0.043	0.070	0.065	0.111
1200	0.017	0.030	0.023	0.037	0.026	0.044	0.040	0.065	0.049	0.078	0.075	0.124
1400	0.019	0.033	0.025	0.040	0.029	0.048	0.045	0.072	0.055	0.086	0.085	0.136
1600	0.021	0.035	0.028	0.043	0.032	0.052	0.050	0.078	0.061	0.093	0.094	0.147
1800	0.024	0.038	0.031	0.046	0.035	0.055	0.055	0.084	0.067	0.100	0.103	0.157
2000	0.026	0.040	0.033	0.049	0.038	0.058	0.060	0.090	0.072	0.107	0.112	0.168

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3.6 Width. For Government procurements only, the width of the finished cloth shall be as specified in the contract or procurement documents.

3.7 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with Title 16 CFR Part 303, Rules and Regulations Under the Textile Fiber Products Identification Act.

3.8 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.3 and 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1).
b. Conformance inspection (see 4.3.2).

4.2 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3.2 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract or purchase order, except where otherwise indicated (see 6.2).

4.3.3 Lot size. For purposes of quality inspection, a lot size shall be no more than 2,500 pounds of material of the same form or dimensions or as specified in the contract. The material shall be produced in one plant under essentially the same conditions and offered for delivery at one time.

4.3.31 Cut and molded items. The lot size shall be the number of cut or molded items in the lot.

4.3.3.2 Sheets and strips. The lot size shall be the number of unit areas in all sheets or strips of the lot. A unit area is defined as an area of one (1) square foot; thus a sheet of two (2) feet wide and 20 feet long would be 40 units, and a strip of four (4) inches wide by 20 feet long would be 7 units.

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4.4 Visual examination. Each lot in the sample shall be examined for defects in accordance with ARPM MO-1. In addition, samples shall be examined for tackiness and brittleness. Any sample sheet, strip, cut or molded part found not to be in accordance with this specification shall not be offered for delivery. If the number of nonconforming items exceeds the acceptable quality limits stated in the contract, it shall be cause for rejection of the lot represented by the sample.

4.5 End item testing. The material shall be tested for the characteristics listed in Table III. The methods of testing as specified wherever applicable and as listed in Table III shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be two (2) continuous yards full width of the finished impregnated cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. Material tests.

Characteristic	Requirement Reference	Test Method
Fiber identification	See 3.4.1	AATCC TM20 (see 6.5)
Fiber content	See 3.4.1	AATCC TM20A (see 6.5)
Weave	See 3.4.2	Visual
Color	See 3.5	See 4.6.1
Weight	Table I	ASTM D3776/D3776M (Opt. C)
Yarns per inch	Table I	ASTM D3775
Breaking strength	Table I	ASTM D5034
Thickness	Table I	ASTM D1777
Width	Table I	Calibrated measuring ruler
Density	Table I	ASTM D792 1/
Fungus	Table I	AATCC TM30, Test III
Permanent set	Table I	See 4.6.2
Oil Resistance: Change in volume (swell) Delamination	Table I	 ASTM D471 2/ 3/
Load deflection	Table II	See 4.6.3

1/ Density in pounds per cubic foot equals specific gravity times 0.03613 times 1728.
2/ The specimens shall be immersed in Reference Fuel F (diesel) at 73 (± 3)°F for a period of 24 hrs (± 1/4) hours.
3/ The specimens tested for change in volume (swell) shall be examined for delamination.

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4.6 Methods of testing and inspection.

4.6.1 Permanent set test. The permanent set test shall be conducted as follows:

Two (2) specimens of the finished coated material shall be cut 2-inch by 2-inch in size. The specimen shall be compressed, perpendicular to the direction of the lamination, between two (2) steel plates under a preconditioning load of 50 lb/in² for 5 minutes. This shall be designated as the "original" thickness.

The load shall be increased at a rate of 500 pounds per minute up to 500 lb/in². The total load shall then be released. The loss in thickness shall be measured between 30 seconds and one (1) minute after the load is removed and expressed as a percentage of the "original" thickness. The test shall be repeated for the same specimen for each specified load (see Table I) within 5 minutes except that no preconditioning load shall be applied. The loss of thickness after each load exposure shall again be determined as a percentage of the "original" thickness. This procedure shall be conducted for each specimen and the average of the two determinations shall be reported as the permanent set after the specified loads.

4.6.2 Load deflection test. The load deflection shall be determined as follows:

Two (2) specimens of the finished coated material shall be cut 2-inch by 2-inch. The specimen shall be compressed, perpendicular to the direction of the lamination, between two (2) steel plates which are held rigidly parallel. The origin of deflection measurements shall be taken at a stress of 5 pounds per square inch (lb/inch²) on the specimen.

The load shall be increased at a rate of 500 pounds per minute and the deflection recorded at each specified load (see Table II). The average material deflection of two (2) specimens shall be reported as the deflection at each specified load.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The laminated cotton or cotton-polyester blend duck cloth covered by this specification is intended for use in vibration attenuation and shock damping.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of material required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2).
d. When first article is required (see 3.1)
e. Color of cloth required (see 3.5).
e. Width of cloth required (see 3.6).
f. Inspection conditions (see 4.3)
g. Conformance inspection (see 4.3.2).
h. Packaging (see 5.1).

6.3 Standard sample. For access to standard samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposals.

6.4 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens".

6.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.6 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.7 Subject term (key word) listing.

Density
Fungus resistant
Load deflection
Permanent set

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